THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is an insulator? 1

(a) iron **(b)** aluminium **(c)** plastic **(d)** graphite

2. The safety device which melts and breaks the circuit when current is too high is a: 1

(a) bulb **(b)** fuse **(c)** switch **(d)** cell

3. A circuit with a gap or a switch in the 'off' position is: 1

(a) a series circuit **(b)** a closed circuit
(c) a parallel circuit **(d)** an open circuit

4. Which non-metal conducts electricity? 1

(a) sulphur **(b)** rubber **(c)** graphite **(d)** wood

5. A fuse wire is made of a material with a: 1

(a) low melting point **(b)** high melting point
(c) high density **(d)** no melting point

6. The device used to open or close a circuit is a: 1

(a) wire **(b)** cell **(c)** switch **(d)** bulb

7. Electric wires are coated with plastic because plastic: 1

(a) is a conductor **(b)** is an insulator **(c)** is heavy **(d)** is shiny

8. A device which switches off automatically when current is too high is an: 1

(a) cell **(b)** MCB **(c)** switch only **(d)** LED

9. The thin wire inside a bulb that glows is the: 1

(a) cap **(b)** fuse **(c)** filament **(d)** terminal

10. The symbol of a cell in a circuit diagram has: 1

(a) a zigzag line **(b)** two equal lines
(c) a long line and a short thick line **(d)** a circle with a cross

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a switch? Why is it used? 2

12. What is the heating effect of electric current? 2

13. Name two conductors and two insulators. 2

14. Why is the filament of a bulb made of tungsten? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw (describe) a circuit diagram with a cell, a bulb and a switch and say how to light the bulb. 3

16. Write three applications of the heating effect of current. 3

17. Give three examples where the heating effect of current is unwanted. 3

OR

List three precautions to be taken while using electrical appliances. 3

18. What are the differences between a fuse and an MCB? 3

OR

Why is a bulb connected to a battery brighter than to a single cell? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the heating effect of current and its applications and disadvantages. 5

OR

Describe the construction of an electric bulb and why the filament glows. 5

20. Describe the electrical safety measures at home. 5

OR

Explain the causes of short circuit and overloading and how to prevent them. 5

END OF QUESTION PAPER — Total: 40 Marks

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